

Pilot Assignment in Cell-Free Massive MIMO

System Model, Pilot Contamination, and Adaptive Assignment

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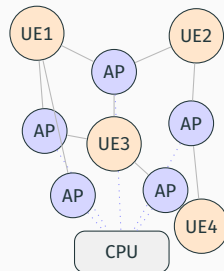
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Basic idea

- Many distributed APs connect to a central processing unit.
- From the UE side, there is no fixed cell boundary.
- A UE can be served by several APs.
- An AP can receive signals from several UEs.

Implication

- Macro-diversity improves coverage.
- Many AP-UE channel links must be estimated.
- Pilot assignment becomes a network-level problem.



Channel between AP l and UE k

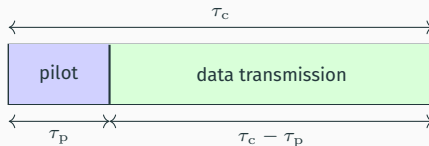
$$\mathbf{h}_{lk} = \sqrt{\beta_{lk}} \mathbf{g}_{lk}, \quad \mathbf{g}_{lk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$$

- L APs, K UEs, N antennas per AP.
- β_{lk} captures pathloss and shadowing.
- Dominant APs/beams are the useful structure for pilot assignment.

Spectral efficiency form

$$\text{SE}_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

Coherence block



Key tension

More pilot symbols can reduce reuse pressure, but they leave fewer symbols for data.

Pilot observation at AP l

$$\begin{aligned}\mathbf{Y}_l^p &= \sum_{i=1}^K \sqrt{p_i} \mathbf{h}_{li} \phi_{t_i}^T + \mathbf{N}_l, \\ \mathbf{y}_{lk}^p &= \mathbf{Y}_l^p \phi_{t_k}^* \\ &= \sqrt{p_k} \tau_p \mathbf{h}_{lk} + \sum_{i \in \mathcal{P}_k \setminus \{k\}} \sqrt{p_i} \tau_p \mathbf{h}_{li} + \mathbf{n}_{lk}. \\ \mathcal{P}_k &= \{i : t_i = t_k\}\end{aligned}$$

Interpretation

- UE k transmits pilot ϕ_{t_k} .
- AP l correlates the received signal with that pilot.
- If another UE uses the same pilot, its channel enters the same statistic.
- The estimate is then contaminated before data detection starts.

Pilot contamination

Same pilot plus overlapping strong channels creates mixed channel estimates.

Why assignment is difficult

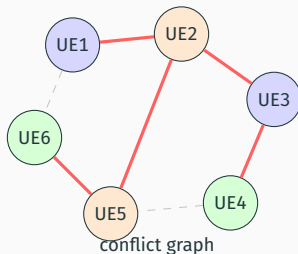
- Orthogonal pilot count is limited: usually $\tau_p < K$.
- Pilot reuse is therefore necessary.
- Each UE can overlap with many other UEs across APs and beams.
- A high-overlap same-pilot pair lowers channel-estimation quality and SINR.

Graph view

- Node = UE.
- Edge = strong effective conflict.
- Color = pilot index.

Research question

Which UEs can reuse a pilot without creating a strong effective conflict?



fewer colors mean fewer actual pilots

$$SE_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

More pilots

- More orthogonal pilot resources.
- Lower pilot reuse pressure.
- Higher training overhead.
- Less data time in the coherence block.

Fewer pilots

- Larger prelog factor.
- Better data-time efficiency.
- Higher contamination risk.
- Works only if the retained structure keeps SINR strong.

Main lens

A useful assignment should reduce pilot overhead without throwing away too much channel quality.

Gao et al.

- AP-domain pilot assignment.
- Many-to-many matching idea.
- Strong small-pilot regime result.
- Pilot count is a fixed design input.

Mussbah et al.

- Beam-domain active/moderate beam graph.
- Conflict graph colored by DSATUR.
- Actual pilot count can become adaptive.
- RF-chain activity matters for EE.

Family	Domain	Conflict signal	Pilot-count rule
Gao	AP / large-scale fading	UE-AP matching	fixed design budget
Mussbah	beam domain	active/moderate beam overlap	graph chromatic number
Proposed set	AP + beam variants	sparsified effective conflicts	adaptive where possible

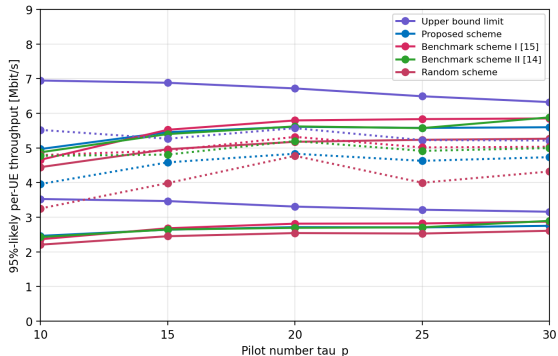
What this slide supports

- 200 Monte Carlo setups.
- $\tau_p = 10$, max-min power.
- Gao vs Random: **+21.7%**.
- Paper claim: approximately +23%.

How to use it

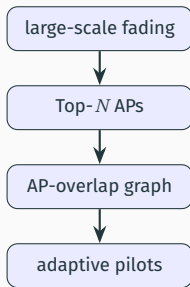
- Use as a reproduction anchor.
- Do not use it as the final multi-antenna benchmark.

Environment: Gao reproduction setting; different from the same-environment multi-antenna results.



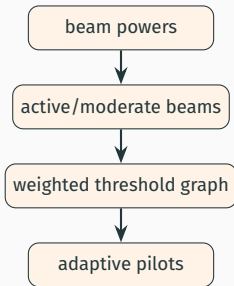
AP-Top-N Adaptive

- Strongest N_{top} APs per UE.
- Conflict by shared Top- N APs.
- Adaptive graph coloring.



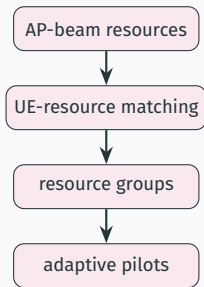
Beam-Weighted Threshold

- Active/moderate beam split.
- Higher weight for active-active overlap.
- Threshold weak conflicts.



Beam-Resource Matching

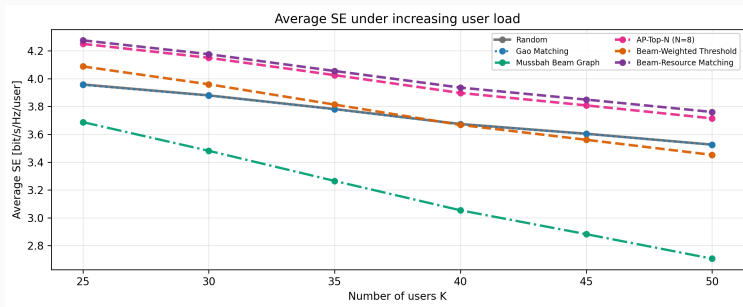
- Candidate AP-beam resources.
- Many-to-many UE-resource matching.
- Adaptive pilot grouping.



Method	Domain	Conflict signal	Pilot count	Role
Random	AP	none	fixed	reference
Gao Matching	AP	UE-AP matching	fixed	reference
Mussbah Beam Graph	beam	binary beam overlap	adaptive	reference
AP-Top-N Adaptive	AP	Top- N AP overlap	adaptive	proposed
Beam-Weighted Threshold	beam	weighted beam overlap	adaptive	proposed
Beam-Resource Matching	beam	AP-beam resource matching	adaptive	proposed

Unifying view

Different conflict signals, same lever: reduce effective conflicts while keeping dominant channel structure.



Environment: $K = 25, \dots, 50$, $L = 200$, $N = 8$, $\tau_c = 150$,
 $\tau_p^{design} = 15$, 200 setups.

Result

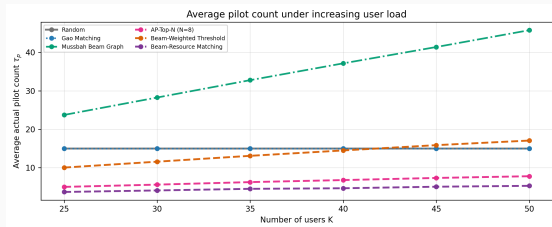
- Gao stays near Random.
- Top-N and Resource stay above Random through $K = 50$.
- Beam-Weighted crosses below Random from $K = 40$.
- At $K = 50$: Resource +6.66%, Top-N +5.35%, Mussbah -23.24%.

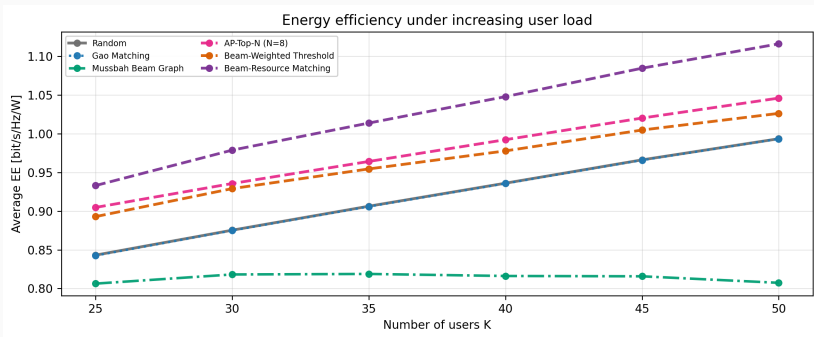
Same-environment K-sweep

$K = 25, \dots, 50$, $L = 200$, $N = 8$, $\tau_c = 150$,
 $\tau_p^{design} = 15$, 200 setups.

Reading

- Random and Gao stay fixed at $\tau_p = 15$.
- Mussbah grows from 23.7 to 45.8 pilots as load increases.
- AP-Top-N and Beam-Resource stay below the design pilot budget.
- Beam-Weighted crosses the budget under heavier load.





Environment: same K -sweep as the SE slide; EE is a proxy using the same selected method set.

Result

- Beam-Resource Matching stays highest across the sweep.
- At $K = 50$: Resource +12.35%, AP-Top-N +5.29%.
- Mussbah loses EE under load as its pilot count grows.

Pilot assignment is a balance problem.
Prune enough to reduce pilots, but not enough to destroy SINR.

What we can defend

- Beam-Resource and AP-Top-N keep positive SE and EE margins.
- Beam-Weighted crossover shows pilot-budget control matters.
- Gao is a reproduction anchor, not the main winner here.

What we should not oversell

- Fewer pilots are not always better.
- $\tau_p = 3$ stress is diagnostic.
- SE gain is not direct evidence of higher SINR.
- EE is a simplified proxy.

Final claim: adaptive pilot assignment is useful when it cuts pilot overhead without visibly sacrificing SINR.

Backup: Graph Consistency Rules

- 1 Use same-environment MC figures for the main ranking.
- 2 Use method names, not person-specific labels.
- 3 Use only Random, Gao Matching, and Mussbah Beam Graph as main references.
- 4 Use Gao reproduction only as a small-pilot reproduction anchor.
- 5 Say “EE proxy” for the current EE plots.
- 6 Claim crossover only for the methods that actually cross in the final six-method plot.
- 7 Add bootstrap CI before using significance language.